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Rock Piles as Kōura Habitat

Stone pile refuges in an Aotearoa New Zealand lake: early benefits for threatened crayfish but not its invasive predator

Olivier V. Raven Ian A. K. Kusabs Robin Holmes
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Stone pile construction at Lake Rotoiti

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Abstract

1. Kōura (*Paranephrops planifrons*), Aotearoa New Zealand’s endemic freshwater crayfish, have declined by up to 91% in Lake Rotoiti (Te Arawa Lakes, Rotorua), driven by predation from the invasive brown bullhead catfish (*Ameiurus nebulosus*) and thermal stratification that compress suitable habitat into the shallow littoral zone. Because these stressors cannot be removed at the lake scale, adding structural refuge may be one of few practical management options.
2. We tested whether stone pile structures deployed in the littoral zone of Lake Rotoiti could provide structural refuge for kōura from predation. Kōura and catfish were monitored at 20 locations (five stone piles, five control, and five paired natural rocky reference sites) for 18 months using a quasi before-after control-impact (quasi-BACI) design, with kōura occupancy modelled using a Bayesian framework that incorporated detection probabilities from baited traps and fyke nets.
3. Stone pile construction produced no statistically significant change in kōura abundance, biomass, or occupancy relative to controls, but effects consistently trended positive across models. Kōura use of stone piles was strongly seasonal: catch rates at stone pile and control sites fell in winter as kōura withdrew to deeper water, while reference sites remained stable. Catfish showed no comparable attraction to stone piles in any season or model.
4. A simulation-based power analysis revealed a promising, developing trend, despite showing the study was underpowered to detect the observed effect size, given high pre-construction kōura occupancy (81–89%) at all site types and only five site pairs monitored.
5. *Synthesis and applications.* Stone piles show early, consistent evidence of benefiting a threatened native crayfish without benefiting its invasive predator and are most valuable when habitat compression and predation pressure peak together in summer and autumn, conditions expected to intensify under climate warming. This study establishes a tested, scalable construction and consenting methodology that lake managers can deploy as an interim refuge-based tool while catfish eradication remains unachievable at the lake scale; we recommend continued monitoring beyond three years and trialling connected, corridor-style structures rather than isolated piles.

1 Introduction

2 Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater
3 fauna now threatened with extinction (Sayer et al., 2025). Among the most endangered are
4 freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss,
5 invasive species, and declining water quality (Richman et al., 2015). As keystone species and
6 ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and food web
7 dynamics, and their loss can have cascading consequences for freshwater ecosystems (Collier
8 et al., 1997; Momot, 1995; Parkyn et al., 2001). In Aotearoa New Zealand, these pressures
9 converge on kōura (*Paranephrops planifrons* White, 1842), an endemic freshwater crayfish now
10 facing rapid population decline in lakes that have long supported some of the largest customary
11 kōura fisheries in the country (Hiroa, 1921; Kusabs & Quinn, 2009).

12 Kōura populations in Lake Rotoiti, located in the Rotorua Te Arawa Lakes region of the North
13 Island of Aotearoa New Zealand (hereafter Aotearoa), have declined by up to 91% over the
14 past two decades (Kusabs et al., 2026). This collapse has virtually ended customary harvest
15 practices for Te Arawa iwi (Māori tribal group) who are hunga tiaki (traditional custodians)
16 of the lake (Kusabs, Hicks, et al., 2015; Kusabs et al., 2026). The decline in kōura numbers
17 is driven by multiple interacting stressors that compound over time in different parts of the
18 lake (Kusabs et al., 2026; Lee et al., 2025). The invasive brown bullhead catfish (*Ameiurus*
19 *nebulosus* (Lesueur, 1819); hereafter catfish), confirmed in Lake Rotoiti in 2016 (Francis, 2019;
20 Grayling, 2020), preys directly on kōura and suppresses their foraging through non-consumptive
21 ‘fear’ effects (Raven, MacNeil, et al., 2026). Predation pressure from catfish is compounded by
22 additional stressors that constrain where in the lake kōura can seek refuge from this invasive
23 predator (Kusabs et al., 2026).

24 During the day, kōura shelter in complex rocky habitat and in deeper water to avoid predation
25 (Jowett et al., 2008; Kusabs, Quinn, et al., 2015), moving out of both at night to forage in
26 the productive littoral zone (Devcich, 1979). The importance of deep-water refuge is shown
27 in Lake Taupō, where kōura persist at greater depth even in the presence of catfish because
28 the lake remains oxygenated below the thermocline (Kusabs & Taiaroa, 2015; Verburg &
29 Albert, 2019). In Lake Rotoiti, this deep refuge becomes anoxic during summer stratification,
30 compressing the zone of liveable habitat for kōura into the shallow littoral margins (Kusabs
31 et al., 2026), precisely where catfish are most active during summer (Blanchfield et al., 2026;
32 Dedual, 2002). Dense beds of invasive macrophytes further restrict kōura movement through
33 this compressed zone and intensify oxygen depletion during die-off events (Hamilton et al.,

1 2005; Kusabs & Quinn, 2009; Vincent et al., 1984). The dual pressures of catfish predation and
2 thermal stratification are expected to only strengthen with predicted climate scenarios (Lee et
3 al., 2025; Woolway et al., 2021), making kōura habitat compression and predation pressure
4 progressively more severe.

5 The compression of suitable habitat and subsequent concentration of kōura into the littoral
6 zone makes the quality of nearshore lacustrine habitats increasingly important for remaining
7 populations. In stream environments, kōura are positively associated with structurally complex
8 habitats including bank undercuts, woody debris, leaf litter, and coarse substrates (Jowett
9 et al., 2008; Parkyn et al., 2009; Parkyn & Collier, 2004; Usio & Townsend, 2000). In lake
10 environments, benthic substrate characteristics are a stronger predictor of kōura distribution
11 than nutrient enrichment or predator abundance (Kusabs, Quinn, et al., 2015), and kōura are
12 positively associated with coarse substrates and riparian vegetation (Raven, Burdon, et al.,
13 2026). Coarse rocky substrate provides refugia from predators and shelter during moulting
14 (Streissl & Hödl, 2002). Experimental studies confirm that shelter availability directly reduces
15 catfish predation on crayfish (Adams, 2007; Zhao et al., 2024), and kōura increase refuge use in
16 the presence of catfish compared to native predators (Raven, MacNeil, et al., 2026), emphasising
17 the importance of shelter availability for kōura populations. Shelter availability is likely a
18 limiting factor for kōura in Lake Rotoiti, where the majority of the shoreline is sandy with
19 dense beds of introduced macrophytes and little rocky structure. Adding rocky substrate to
20 soft-bottomed lake areas has been shown to significantly increase crayfish abundance (Johnsen
21 & Taugbøl, 2008), and increasing structural complexity in streams improves crayfish restocking
22 success (Huolila et al., 1997). Deploying habitat enhancement structures that add structural
23 complexity in the littoral zone could potentially be a practical way to restore kōura populations
24 in Lake Rotoiti.

25 In this study habitat enhancement structures in the shape of stone piles were deployed along
26 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
27 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
28 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
29 kōura occupancy, abundance and biomass would increase more rapidly at the stone pile sites
30 than at control sites, as intense catfish predation pressure in the littoral zone was expected
31 to compel kōura actively toward available refuge structures. We further predicted that kōura
32 abundance would continue to increase progressively after construction, at least within the first
33 18 months, and that catfish would instead show seasonal patterns of shoreline use rather than
34 a construction-driven trend (Barnes, 1996; Blanchfield et al., 2026; Dedual, 2002). Finally, we

1 predicted that stone pile sites would progressively converge toward kōura densities observed at
2 natural rocky reference sites, and that catfish would not exhibit a similar pattern.

3 **Materials and Methods**

4 **Study location and construction**

5 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua Te
6 Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m). The lake
7 stratifies for approximately nine months a year (**?@fig-lake-conditions**), with pronounced
8 thermal stratification compressing oxygenated habitat into the shallow littoral zone during
9 summer and autumn. Since 2008, nutrient-rich inflows from Lake Rotorua have been diverted
10 toward the Kaituna River via the Ōhau Channel diversion wall to improve water quality in
11 Lake Rotoiti ([Hamilton et al., 2005](#); [Prentice et al., 2026](#)). Lake level is regulated by a weir at
12 the outlet of the Kaituna River, providing a stable and controlled water level throughout the
13 year. Catfish are present throughout the lake but occur in particularly high concentrations
14 in Te Weta Bay on the Northwest side of the lake ([Francis, 2019](#)). Five stone pile sites were
15 selected alongside the lake shoreline based on accessibility and landowner approval (Figure 1).
16 At each stone pile site, a paired control site was established 50 m along the shoreline, matched
17 for habitat characteristics e.g., water depth, substrate type and vegetation cover. Five naturally
18 occurring rocky shoreline locations, with each two monitoring sites 50 m apart were selected
19 as reference. In total, 20 sites of each 150 m² (15 × 10 m) were monitored seasonally over 18
20 months. Physical characteristics of all sites are summarised in Table 1. Baseline monitoring was
21 conducted in February 2025 prior to stone pile deployment, which occurred the following week.
22 Stone piles were constructed from locally sourced quarry rock (Manawahe Stone and Quarry,
23 ~25 km from the lake) with a diameter ranging from 85 - 315 mm (mean = 173 mm). At each
24 of the five stone pile sites, three stone piles were arranged horizontally along the shoreline by
25 hand from the shore. Stone pile design was informed by mesocosm experiments demonstrating
26 kōura preference for enclosed stone structures (unpublished data). Stone piles were positioned
27 approximately 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².



Figure 1: Map of Lake Rotoiti in Bay of Plenty region of New Zealand. The monitoring sites are coloured by site type: stone pile sites red, control site yellow, and reference sites blue. Ten close-up maps show the placement of each of the sites.

Table 1: Physical characteristics of stone pile, control, and reference monitoring sites at baseline. Sites 1–10 are stone pile (odd) and paired control (even) sites; sites 11–20 are reference sites. Substrate index is a weighted composite of substrate particle sizes (higher values indicate coarser substrate more suitable for kōura). Slope is the gradient to the 20 m depth contour. Fetch is maximum effective fetch. Vegetation values are visually estimated percentage cover.

Site	Type	Habitat	Substrate index	Slope (m/m)	Fetch (km)	Riparian veg. (%)	Overhang (%)	Wood cover (%)
1	Stone pile	Rocky	5.20	0.12	4.76	0	10	0
2	Control	Rocky	4.40	0.11	4.79	70	0	0
3	Stone pile	Sandy	3.00	0.01	1.71	100	0	0
4	Control	Sandy	3.00	0.01	1.74	80	5	0
5	Stone pile	Sandy	2.40	0.01	0.61	10	0	0
6	Control	Sandy	3.00	0.01	0.55	0	0	0
7	Stone pile	Sandy	3.00	0.06	2.50	100	0	0
8	Control	Sandy	3.00	0.05	2.22	30	0	0
9	Stone pile	Sandy	3.35	0.08	3.59	0	0	3
10	Control	Rocky	5.00	0.09	3.74	20	0	0
11	Reference	Rocky	6.60	0.16	4.47	0	0	0
12	Reference	Rocky	6.20	0.12	4.17	100	100	0
13	Reference	Sandy	3.60	0.09	1.71	100	100	0
14	Reference	Sandy	3.50	0.11	1.65	100	100	0
15	Reference	Rocky	5.20	0.03	1.44	100	100	6
16	Reference	Rocky	5.20	0.04	1.40	100	100	2
17	Reference	Rocky	5.00	0.05	3.47	100	100	30
18	Reference	Rocky	6.20	0.05	3.46	100	100	7
19	Reference	Rocky	5.70	0.11	2.63	100	100	4
20	Reference	Rocky	5.70	0.13	2.87	100	100	0

¹ Source: [Article Notebook](#)

2 Monitoring

3 Physical, chemical, and biological parameters were recorded at each site during each monitoring
4 round. Static site characteristics including substrate composition, slope to depth contour,
5 riparian vegetation, overhanging tree cover, and wood cover were visually estimated during
6 baseline monitoring only, as these were not expected to change substantially over the monitoring
7 period. Substrate composition was visually estimated by a single observer walking around
8 each site and using an underwater viewing chamber, categorising bedrock, boulders (>256
9 mm), cobble (64–256 mm), gravel (2–64 mm), sand (0.063–2 mm), mud (<0.063 mm), and

1 organic matter ([Wentworth, 1922](#)). A substrate index, that indicates suitability for kōura, was
2 calculated from these estimates ([Jowett et al., 2008](#); [Jowett & Richardson, 1990](#)):

$$\begin{aligned}\text{Substrate index} = & 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} \\ & + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} \\ & + 0.02 \times \text{organic matter} + 0.01 \times \text{mud}\end{aligned}$$

3 Coarser substrates were assigned higher weights as they provide greater interstitial pore space
4 and stable structures used by kōura ([Nyström et al., 2006](#); [Usio & Townsend, 2000](#)).

5 After stone pile construction, measurements of the size and dimensions of each structure were
6 taken. The post-construction substrate index for stone pile sites was recalculated using each
7 site's combined stone-pile footprint area, expressed as a percentage of the 150 m² site area
8 and treated as 80% cobble and 20% boulders displacing an equivalent area of sand, to reflect
9 their contribution to substrate complexity. Slope was assessed as the horizontal distance from
10 the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a
11 projected coordinate system (NZTM2000; EPSG:2193).

12 Seasonal variable parameters were recorded at every monitoring round and included aquatic
13 vegetation cover (emergent and submerged macrophytes). Chemical parameters included water
14 temperature (°C), dissolved oxygen (mg/L), and conductivity (S/cm) were measured using a
15 YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured using a
16 handheld pH meter (EC-PCTestr35, Eutech Instruments).

17 Kōura and fish were sampled every monitoring round using two single-winged, double-throated
18 fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250
19 mm; two 50 mm entrance openings) baited with catfish flesh ([Joy et al., 2013](#); [MacNeil et](#)
20 [al., 2025](#)). Fyke nets were placed perpendicular to the shoreline outside of the stone pile
21 structures or at 10 m intervals, with bait traps positioned between them. All equipment was
22 deployed simultaneously at each site for a 24-h period. Captured kōura were measured for
23 orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet
24 mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC,
25 Australia). Catfish and other fish species, were recorded for abundance, measured for total
26 length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort
27 (BPUE) were calculated for all kōura and fish based on the number of nets and traps deployed at
28 each site per monitoring round, standardised by soak time ([Kusabs, Quinn, et al., 2015](#)). Kōura
29 weights were unavailable for 36 individuals because of scale failure and were therefore estimated

1 from a log –log length–weight regression fitted to individuals with both measurements (n
2 = 840), following standard approaches for weight length relationships in aquatic organisms
3 **?@fig-length-weight-ch5** (Froese, 2006). Predicted values were back-transformed with a
4 lognormal bias correction factor $10^{0.5\sigma^2}$ (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.684 \times \log_{10}(\text{OCL length [mm]})$$

5 Data analysis

6 Data analysis was a three-part process, the first assessed baseline equivalence between stone
7 pile, control, and reference sites before construction, the second assessed the impact of stone
8 pile construction and after-construction trajectory on kōura and catfish occupancy, abundance,
9 and biomass, and the third assessed whether stone pile sites converged toward reference sites
10 and if reference sites remained stable over time. All analyses were conducted in R (version
11 4.6.1; R Core Team (2026)) using the glmmTMB and emmeans packages (Brooks et al., 2017;
12 Lenth & Piaskowski, 2017).

13 Covariate selection for all models followed a consistent procedure. Corrected Akaike Information
14 Criterion (AICc) values for each base model were compared against models with additional
15 covariates (temperature, season, substrate index, maximum effective fetch, or wave energy
16 proxy) added at a time, and a covariate was retained only when it reduced AICc by more than
17 two units relative to base model (Burnham & Anderson, 2002). Robustness and sensitivity for
18 all models were assessed by evaluating uniform distribution, dispersion and outlier rate using
19 the DHARMA package (Hartig, 2016). Across the full set of models, the Benjamini–Hochberg
20 correction method was applied to control the false discovery rate within each of the three
21 analysis components.

22 Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank
23 tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum
24 tests to compare stone pile sites with reference sites. This was done for response variables
25 (water quality, substrate, and kōura and catfish CPUE and BPUE) to confirm that the stone
26 pile and control sites were equivalent before construction, and to characterise the difference
27 between stone pile and reference sites. To further assess baseline equivalence, generalised
28 linear models (GLMs) were fitted to test whether kōura and catfish CPUE and BPUE differed
29 between site type (stone pile / control) and between the two reference sites (reference 1 /

1 reference 2) using a Tweedie family with a log link. Following the covariate selection procedure,
2 temperature was included in the catfish models comparing reference sites, as the base model
3 failed to converge and temperature was required for numerical stability.

4 The second component of the analysis tested construction impact relative to control sites using
5 a quasi-before-after, control-impact (quasi-BACI) design ([Underwood, 1994](#)), and separately
6 tested after-construction trajectory of kōura and catfish populations. The design is quasi-
7 BACI because only a single pre-construction monitoring round was possible. The quasi-BACI
8 design compared one pre-construction monitoring round with six post-construction monitoring
9 rounds using generalised linear mixed effects models (GLMMs) to test the effect of stone pile
10 construction on kōura and catfish occupancy, abundance, and biomass. Time period (before
11 / after) and site type (stone pile / control) were included as an interaction term, monitoring
12 round as a covariate and site nested within control-impact pair as a random effect. Occupancy
13 was modelled using a binomial GLMM, and abundance and biomass using a Tweedie GLMM
14 with a log link; fetch was retained as a covariate in the catfish model. The post-construction
15 trajectory models used the same response variables, families, covariate, and random effect
16 structure, but with site type and monitoring round included as an interaction term in place of
17 the before / after comparison.

18 The third component of the analysis assessed whether stone pile sites converged toward reference
19 sites over time, and whether reference sites themselves remained stable. Kōura and catfish
20 abundance and biomass were tested using GLMMs with site type (stone pile / reference) and
21 monitoring round included as an interaction term, and site number as a random effect, using
22 a Tweedie family with log link; temperature was retained as a covariate in the catfish model.
23 Reference site stability was tested in GLMMs with site type (reference 1 / reference 2) and
24 monitoring round as an interaction term, and site nested within pair as a random effect, again
25 using Tweedie family with log link; season was retained as a covariate in the kōura model and
26 temperature was retained as a covariate in the catfish model.

27 To complement the GLMM abundance analysis, kōura site-level occupancy was estimated using
28 a two-method occupancy model fitted within a Bayesian framework. Because bait traps and
29 fyke nets provide independent and simultaneous detection opportunities at each site, true site
30 occupancy could be separated from imperfect and method-specific detection probabilities. A
31 custom Stan observation model was implemented using the brms package ([Bürkner, 2017, 2018](#))
32 using cmdstanr ([Gabry et al., 2025](#)) to estimate true occupancy and detection probabilities
33 for baited traps and fyke nets via logit links. Standardised water temperature was included
34 as the covariate for detection probability. Three occupancy sub-models were fitted. A full

1 model across all monitoring rounds served as baseline comparison. A summer-only model and
2 a warm-season model (summer and autumn combined) were fitted to focus on the period of
3 highest thermal stratification, when kōura are known to disproportionately occupy nearshore
4 littoral habitats (Kusabs et al., 2026) and when catfish predation pressure is highest in the areas
5 where the stone piles are located (Blanchfield et al., 2026). The warm-season model maximises
6 ecological relevance, while retaining a tractable formula with season as an additive covariate.
7 All models were run with four chains of 4000 iterations (1500 warmup) and convergence was
8 assessed using R values (threshold < 1.01) and bulk effective sample sizes (threshold > 400)
9 (Vehtari et al., 2021). Divergent transitions were rare in all models (9 of 10,000 post-warmup
10 draws) and considered negligible given the otherwise strong convergence diagnostics.

11 A simulation-based power analysis was conducted to quantify the sample size required for
12 statistical detection of the BACI effect under the observed effect size and variance structure,
13 and to put the non-significant results in perspective relative to study design rather than true
14 absence of effect. For each analysis the occupancy and abundance, datasets were simulated
15 across a grid of scenarios varying in number of site pairs (3-30) and after-period monitoring
16 rounds (2-12). For the occupancy BACI, simulated binary detection data were generated
17 using values from the warm-season occupancy model posterior means, and each simulated
18 dataset was analysed using a logistic GLMM (lme4; Bates et al. (2015)). For the CPUE BACI,
19 simulated abundance data were generated using fixed effects and variance components from the
20 kōura abundance BACI model, using a log-normal approximation to the Tweedie distribution,
21 and each dataset was analysed using a linear mixed model fitted to a log-transformed CPUE.
22 Statistical power was estimated as the proportion of simulations (300 per scenario) in which
23 the model's 95% confidence interval for the BACI coefficient excluded zero in the positive
24 direction.

25 Results

26 Kōura presence was detected at stone pile sites within three months of construction, with
27 occupancy and abundance trending consistently higher at stone pile sites than paired control
28 sites throughout the 18-month post-construction period, though this difference did not reach
29 statistical significance. Results across all models and response variables are summarised in
30 Figure 2 and ?@tbl-models-summary-full-supp.

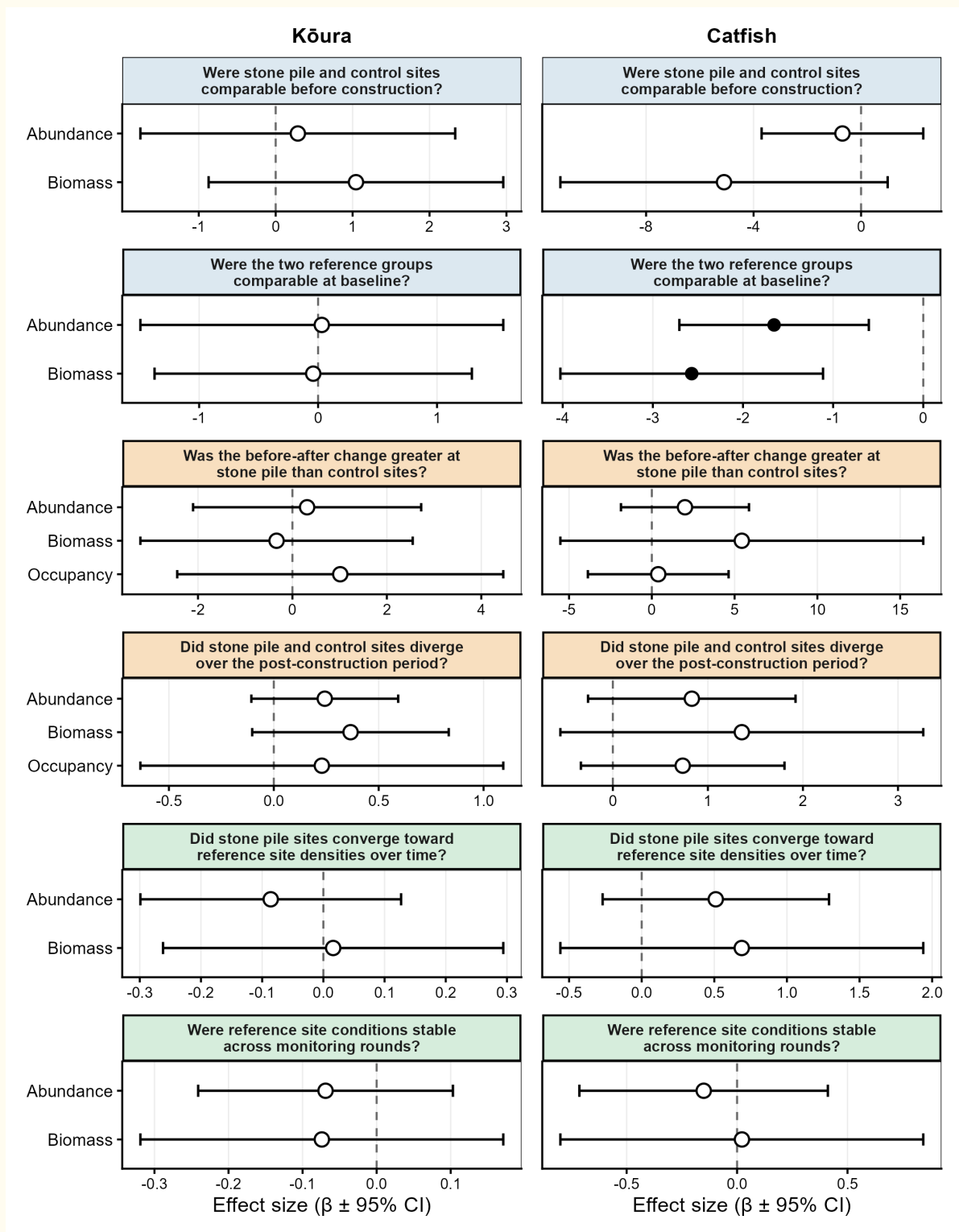


Figure 2: Effect sizes ($\beta \pm 95\% \text{ CI}$) for the key term of interest across all models, separated by species (columns) and comparison type (rows). Filled points indicate significant effects ($p < 0.05$); open points are non-significant. Each row has its own x-axis scale, effect magnitudes are not directly comparable across rows because each comparison tests a different contrast (step-change, per-round slope, or simple group difference), and occupancy is on the log-odds scale while abundance and biomass models use the Tweedie log link. Models with degenerate confidence intervals are excluded.

Baseline equivalence

Pre-construction, kōura abundance and biomass did not differ significantly between stone pile and paired control sites (?@tbl-baseline-combined-suppl), confirming baseline equivalences ($p = 0.783$ and $p = 0.571$). Kōura abundance ($p = 0.968$) and biomass ($p = 0.968$) did not differ between reference sites. Catfish abundance ($p = 0.004$) and biomass ($p = 0.002$) likewise did not differ between stone pile and control sites pre-construction. However, catfish abundance and biomass were significantly higher at Reference 1 than Reference 2 ($p = 0.004$ and $p = 0.002$, respectively), despite both sites having comparable rocky substrate. This suggests catfish distribution varies considerably between nearby sites with similar habitat characteristics.

Construction impact (BACI) and post-construction trajectory

Stone pile construction produced no detectable step-change in kōura abundance, biomass, or occupancy relative to paired control sites over the 18-month monitoring period following deployment (all $p_{\text{adj}} = 0.858$) (Figure 3). A matched-season comparison, comparing Summer 2025 (before construction) with Summer 2026 (one year after construction), produced the same conclusion (kōura abundance: $p = 0.582$). Post-construction trajectories did not diverge significantly between stone pile and control sites (all $p_{\text{adj}} = 0.215$). Kōura abundance declined significantly across post-construction rounds at control and stone pile sites ($\beta = -0.41$, $p = 0.004$), with the sharpest decline observed in Winter 2026 (Figure 4). Stone pile construction produced no detectable step-change or trajectory divergence in catfish abundance, biomass, or occupancy relative to control sites across any model or time period (Figure 2).

Kōura occupancy was high across all site types before construction, 81–89% in summer (?@tbl-psi-warm-suppl). After construction, occupancy increased marginally at all sites, 89–94% in summer, thereby limiting our ability to detect a treatment effect. The BACI contrast to compare how much stone pile sites changed relative to control sites was slightly positive, but the credible intervals spanned zero with wide uncertainty (Figure 5), indicating no detectable additional increase at stone pile sites. However, this result likely reflects a measured ceiling rather than biological response. After-treatment contrast showed no significant difference between site types, but the direction of the effect was consistent with early habitat use: stone pile sites showed a positive contrast relative to control sites, while control sites showed a negative contrast relative to reference sites, and stone pile sites were near zero relative to reference sites (Figure 5).

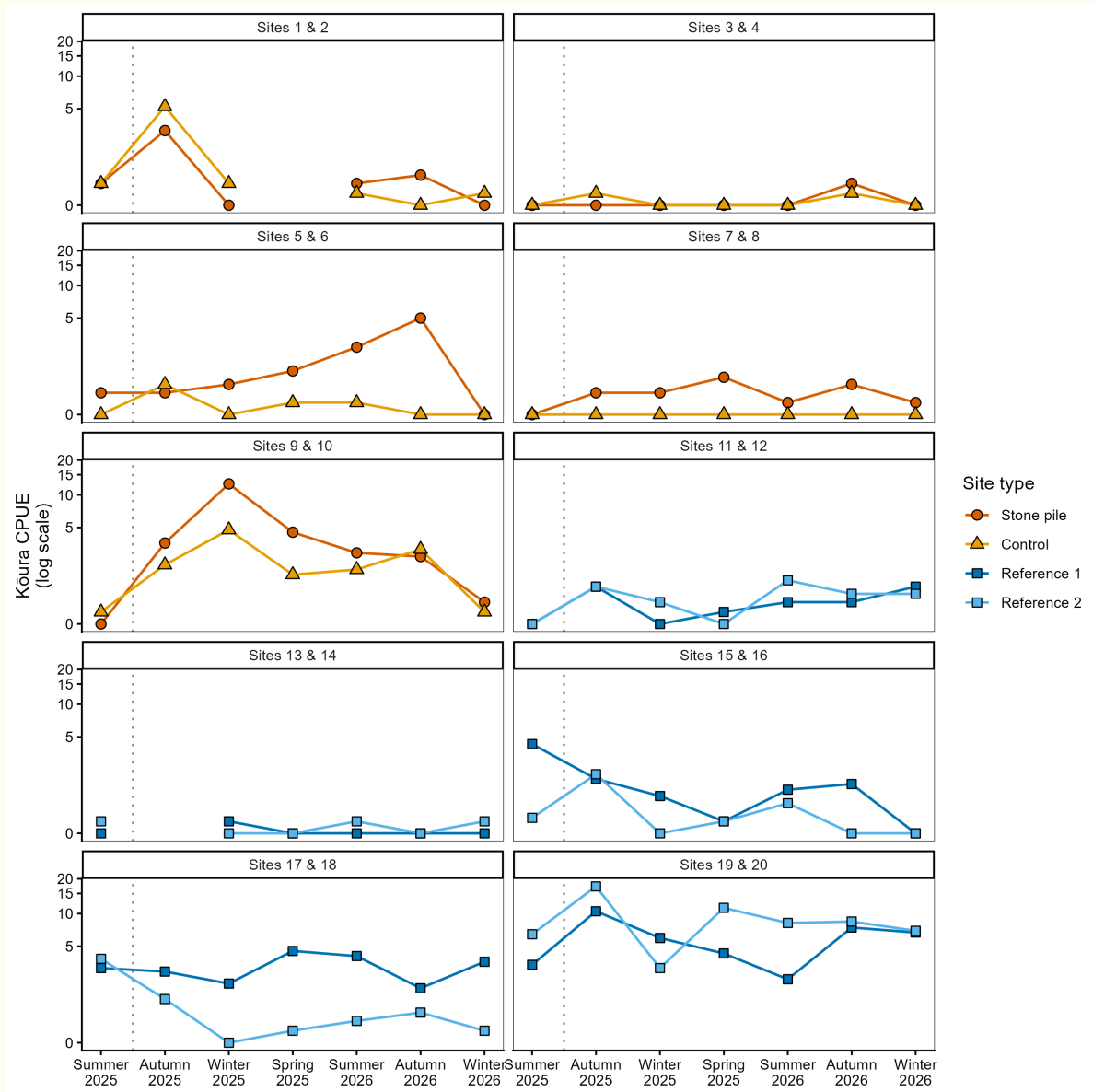


Figure 3: Kōura CPUE at individual site pairs across all monitoring rounds. Each panel shows one stone pile / control pair (sites 1–10) or reference pair (sites 11–20). Colour and shape indicate site type. Vertical dotted line marks stone pile construction. y-axis on $\log(1+x)$ scale.

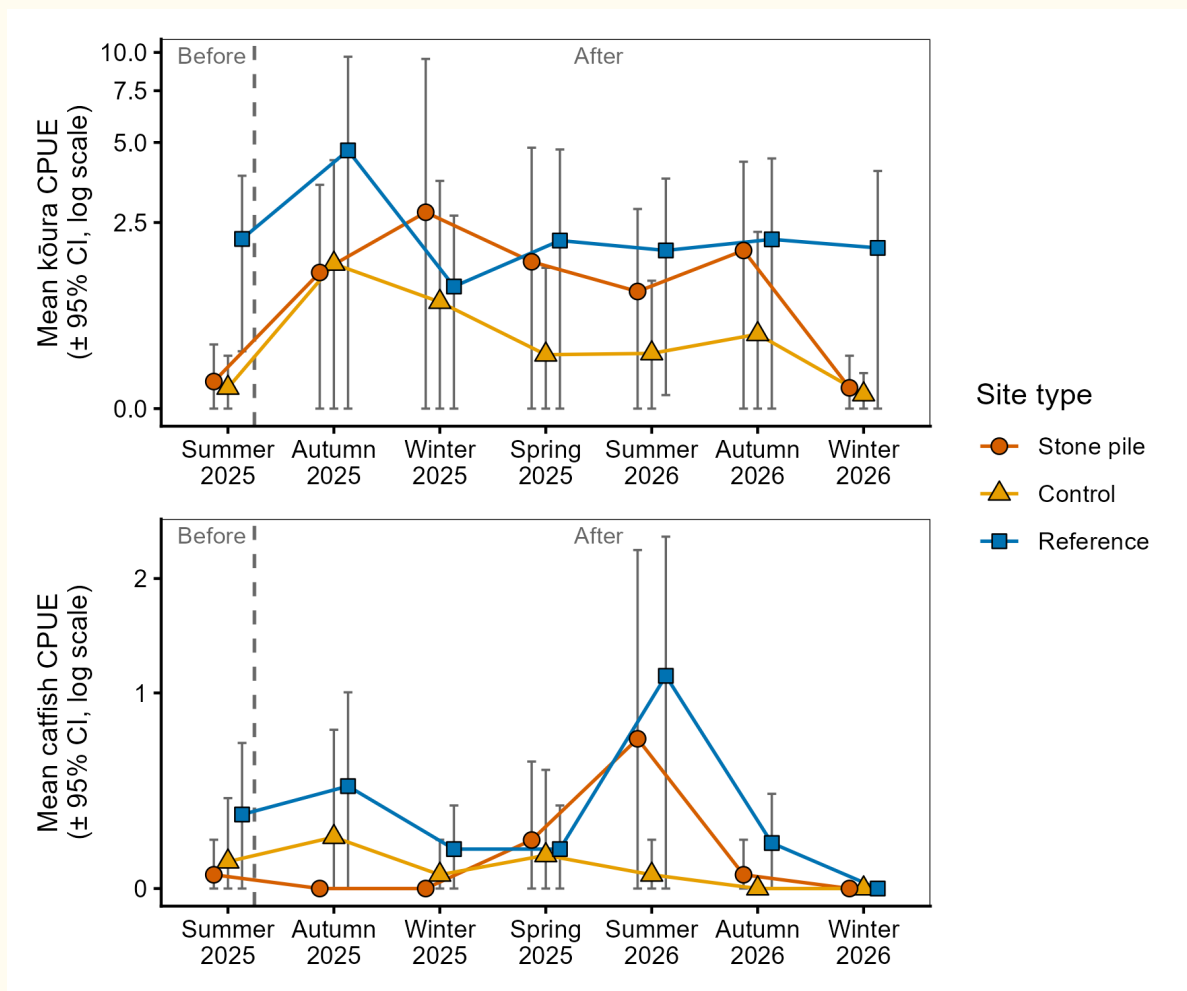


Figure 4: Mean CPUE per monitoring round ($\pm 95\%$ CI, log scale) by site type for kōura (top) and catfish (bottom). Vertical dashed line distinguishes the pre-construction period (Before) from post-construction monitoring (After).

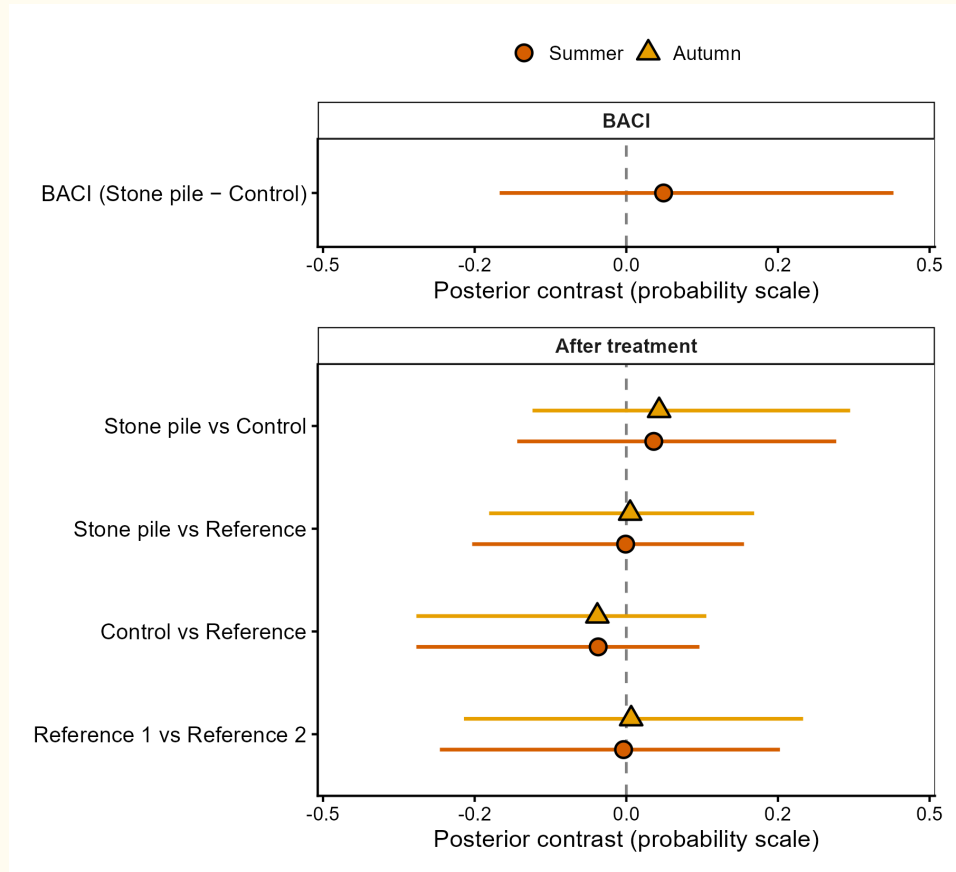


Figure 5: Posterior contrasts in kōura occupancy from the warm-season model. BACI contrast (stone pile minus control, before vs after construction). After-treatment contrasts between all site-type pairs. Points show posterior means with 95% credible intervals; contrasts overlapping zero indicate no detectable difference.

1 The non-significant BACI results should be interpreted in the context of statistical power. For
 2 occupancy, power was near zero across all simulated scenarios (maximum 8%), reflecting the
 3 ceiling effect leaving no statistical room for a BACI signal (?@fig-power-curves-combined).
 4 Post-construction power analysis showed that achieving 80% power with the current five site
 5 pairs would require an effect size of approximately 1.8 log-units, almost 6 times the current
 6 observed effect (Figure 6). If the estimated positive trajectory of +0.24 log-units per seasonal
 7 round continues, the effect size may reach this detection threshold within approximately 6
 8 additional monitoring rounds (~1.5 years).

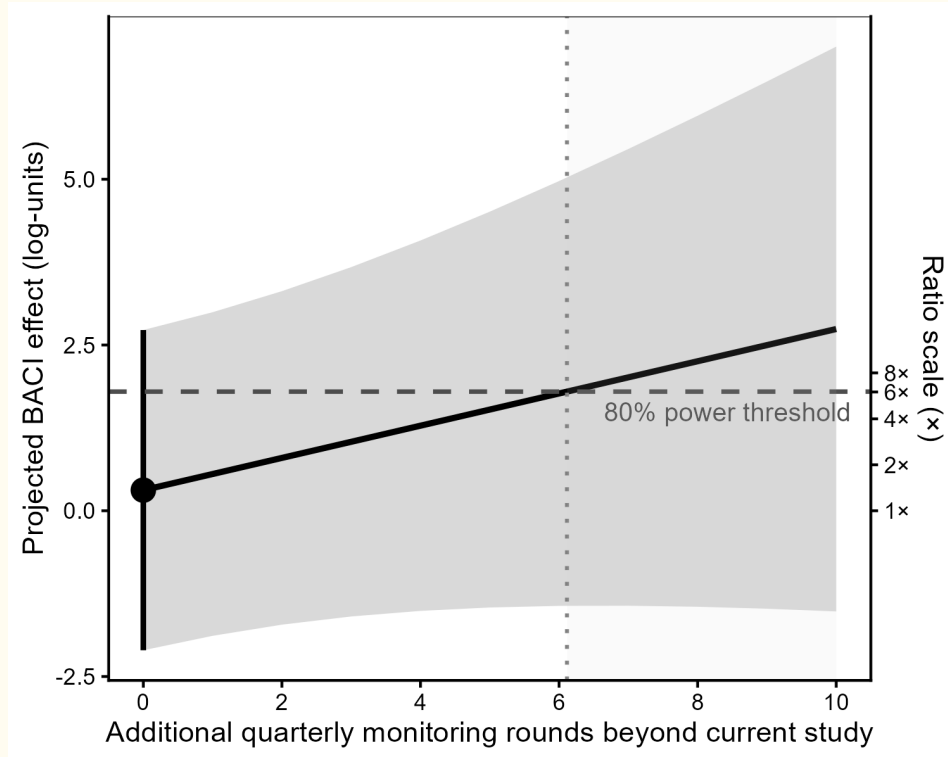


Figure 6: Projected kōura CPUE BACI effect size over additional monitoring rounds. The solid line shows the linear projection based on the estimated post-construction trajectory; shaded ribbon shows 95% uncertainty interval. Dashed horizontal line marks the detection threshold for 80% power with five site pairs.

9 Convergence toward reference sites and reference stability

10 Stone pile sites maintained a consistently lower kōura abundance than reference sites throughout
 11 the post-construction period, with no significant change in this gap over time ($\beta = -0.09$, p

1 = 0.429; ?@fig-convergence-plot). Kōura abundance declined across monitoring rounds at
 2 both stone pile and reference sites ($\beta = -0.09$, $p = 0.107$). The Winter 2026 monitoring round
 3 showed a pronounced decline at stone pile sites (CPUE 0.7) while reference sites maintained
 4 substantially higher abundance (CPUE 1.7; Figure 4), a within-round contrast not captured by
 5 the linear trajectory model. Water temperature at Winter 2026 was approximately 1.5°C lower
 6 than Winter 2025 but consistent across site types, ruling out temperature as a site-specific driver.
 7 Visual inspection of stone pile surfaces during Winter 2026 monitoring revealed substantially
 8 greater filamentous algal accumulation compared to earlier rounds (?@fig-stone-piles). Catfish
 9 abundance and biomass showed no significant convergence or divergence between stone pile
 10 and reference sites over time (Figure 2).

11 Discussion

12 Stone pile structures were deployed in the littoral zone of Lake Rotoiti to test whether physical
 13 refuge could reduce predation pressure on kōura by invasive catfish. Stone piles produced no
 14 statistically significant increase in kōura abundance, biomass, or occupancy relative to paired
 15 control sites, but the direction of effect was consistently positive across every response variable
 16 and model, and occupancy at stone pile sites converged with occupancy at natural rocky
 17 reference sites by the first post-construction summer. Catfish showed no comparable response
 18 to stone pile presence in any model, at any site type, or in any season, indicating that stone
 19 piles selectively benefit kōura without providing equivalent refuge or foraging opportunity for
 20 this invasive predator. A pronounced decline in kōura catch rates at stone pile and control sites
 21 in Winter 2026, evidently driven by seasonal redistribution to deeper water rather than habitat
 22 abandonment, showed that structural refuge value is itself seasonal and concentrated in the
 23 warmer months when catfish predation pressure and habitat compression peak. Together, these
 24 results and a formal power analysis indicate that the absence of statistical significance most
 25 plausibly reflects limited study duration and sample size rather than genuine ineffectiveness of
 26 the intervention.

27 Construction effect and statistical power

28 Kōura catch rates were highly variable across all site types throughout the monitoring period,
 29 with wide confidence intervals, likely reflecting genuine biological variability in kōura distribution.
 30 This variability may reflect seasonal activity patterns, detection probability changes across

1 monitoring rounds, and natural heterogeneity in littoral zone use, rather than stable site
2 associations. Despite this variability, the Bayesian occupancy model revealed a consistent
3 positive directional trend, with stone pile sites showing higher occupancy than control sites,
4 and occupancy comparable to natural rocky reference sites across both summer and autumn
5 monitoring rounds. Whether this reflects active refuge-seeking driven by catfish predation
6 pressure or gradual colonisation and population expansion into newly available habitats cannot
7 be determined from the current data, but the direction of the response was consistent with
8 the purpose of the structures, yet no statistically significant effect was detected within the
9 18-month monitoring period.

10 The absence of a significant effect is most likely explained by a combination of monitoring
11 duration, statistical power, and system dynamics rather than ineffectiveness of the structures.
12 Habitat establishment for freshwater crayfish can be inherently slow. Johnsen & Taugbøl (2008)
13 reported that crayfish abundance at a stone addition site took 13 years to reach reference
14 levels, and monitoring periods shorter than two years are often insufficient to detect significant
15 responses to habitat modification (Pilotto et al., 2018; Ruhí et al., 2012; Testa et al., 2011).
16 Following reconstruction of the Lake Rotorua lakefront, which is connected to Lake Rotoiti,
17 kōura showed no meaningful recolonisation after 28 months (Kusabs, 2023). They attributed
18 slow recolonisation to persistent catfish predation pressure and distance from source population,
19 consistent with time scales observed here. The power analysis indicates that five site pairs
20 are insufficient to reliably detect effects of the magnitude observed here within 18 months,
21 establishing a minimum design threshold: future stone pile monitoring programmes should
22 plan for at least three years of monitoring and expect high pre-construction occupancy, leaving
23 little statistical room for a measurable treatment effect. The simultaneous elevation of kōura
24 CPUE across all site types in Autumn 2025 further complicated detection of an early treatment
25 effect in the BACI contrast.

26 Features of the system itself may also contribute to the weak response. Kōura abundance is
27 highest in lakes without catfish (Raven, Burdon, et al., 2026), suggesting predation pressure
28 may suppress kōura populations well below the carrying capacity of existing habitat. If so,
29 structural habitat availability may not currently be the primary limiting factor, and a weak
30 or delayed response to habitat addition would be expected. Equally, if kōura are already at
31 carrying capacity of existing rocky habitat, stone piles add new capacity that will only be filled
32 gradually as the population expands, which is unlikely to be detectable within 18 months. In
33 either case, a delayed rather than absent population response would be expected. Isolated stone
34 pile structures may also be difficult for kōura to locate, since distance between isolated habitat

1 patches is a key determinant of colonisation success (Hanski, 1998). Each of these explanations
2 point toward a delayed, rather than absent, population response, and continued monitoring
3 would be the appropriate next step rather than a reassessment of the intervention itself.

4 **Seasonal dynamics: the Winter 2026 decline**

5 Kōura CPUE at stone pile and control sites fell sharply in Winter 2026, returning to levels
6 close to the pre-deployment baseline from Summer 2025, while reference sites maintained
7 substantially higher abundance. This pattern is consistent with known seasonal migration
8 behaviour across lake depth profiles and kōura activity rates during cooler water temperatures
9 (Devcich, 1979). When lake thermal stratification breaks down in winter, the hypolimnion
10 re-oxygenates and deep-water habitat becomes accessible, allowing kōura to redistribute along
11 the depth gradient (Devcich, 1979). Reduced activity at low water temperatures independently
12 lowers the probability of trap capture (Devcich, 1979), and although Winter 2026 was 1.5 °C
13 lower than Winter 2025, this difference was consistent across all site types and therefore cannot
14 explain why reference sites were substantially more resilient to the winter decline than the
15 stone pile sites.

16 Natural rocky reference sites in this study represent the ecological target for stone pile deploy-
17 ment. Reference sites had established kōura populations, connectivity to deeper waters, and
18 extensive refuge habitat, allowing kōura to maintain littoral presence through winter or return
19 rapidly from deeper refuges as conditions fluctuated. Stone pile sites are isolated structures with
20 less connectivity to deep and extensive refuge habitat, meaning kōura that withdrew in winter
21 had no quick pathway back to the structures. This difference likely explains why stone pile sites
22 maintained consistently lower kōura abundance than reference sites throughout the monitoring
23 period, with no significant convergence toward reference levels. However, kōura occupancy at
24 stone pile sites was comparable to that at reference sites after construction, suggesting that
25 kōura found and used the structures at similar frequency to natural rocky habitat but at lower
26 densities. This distinction between occupancy and abundance likely reflects the early stage of
27 colonisation and the small structural footprint of stone piles relative to the extensive natural
28 rocky habitat at reference sites. These differences are expected to narrow as biofilm succession
29 matures and kōura accumulate at the structures over time.

30 Extensive filamentous algae growth was observed on the stone pile structures in Winter 2026
31 but not to the same degree in Winter 2025 (?@fig-stone-piles), consistent with the seasonal
32 pattern typical for thermally stratified lakes, where winter mixing raises nutrient availability

1 while cooler water suppresses invertebrate grazing, allowing algal mats to accumulate on hard
2 surfaces ([Bonacina et al., 2023](#); [Wetzel, 2001](#)). The freshly quarried piles were free from biofilm
3 at deployment and by Winter 2026 supported a diverse meiofaunal assemblage, indicating
4 active succession rather than habitat degradation; algal cover would be expected to decline as
5 grazer activity resumes in spring and summer. Despite the near-zero trap catches in Winter
6 2026, night spotlighting at two stone pile sites in August 2026 directly observed kōura sheltering
7 within the structures, confirming that low winter catch rates reflect reduced detectability rather
8 than abandonment. Together, these results indicate that stone pile refuge value is inherently
9 seasonal — greatest in summer and autumn, when thermal stratification compresses habitat
10 into the littoral zone and catfish predation peaks, and lowest in winter, when kōura redistribute
11 to deeper water regardless of structure type.

12 **Catfish: no evidence of attraction to stone piles**

13 Catfish showed their seasonal patterns in littoral zone use, but these were unrelated to stone pile
14 presence. Stone pile construction produced no evidence of catfish attraction or accumulation.
15 Even if catfish were to use stone piles to some degree, the ecological significance of this would
16 be limited given the extensive availability of existing suitable habitat for catfish in Lake
17 Rotoiti, particularly invasive macrophyte beds ([Barnes & Hicks, 2003](#)). The single-site increase
18 in catfish CPUE during Summer 2026 was mirrored by an increase at reference sites and
19 was consistent with lake-wide seasonal redistribution into warm littoral waters rather than
20 any stone-pile-specific event ([Blanchfield et al., 2026](#)). Overall catfish catches remained low
21 throughout the study relative to targeted removal netting in Te Weta Bay, where highest catfish
22 densities are concentrated ([Francis, 2019](#)). In autumn, catfish shift to mid-water movement
23 across the lake basin remaining above 15 m depth ([Blanchfield et al., 2026](#)), reducing predation
24 pressure on kōura in the littoral zone. This absence of attraction is an important management
25 result in its own right: habitat enhancement can be targeted at a threatened native species
26 without inadvertently improving conditions for its invasive predator.

27 **Implications for restoration design and monitoring**

28 Predator-prey interactions are shaped by habitat structure, which offers refuge to prey, influences
29 how easily predators locate and capture prey, and alters the costs and benefits of hunting for
30 a predator ([Lennox et al., 2025](#)). This framework applies directly to catfish-invaded lakes,

1 where habitat addition offers a practical lever on that interaction, and rocky environments have
2 been identified as natural multi-threat refugia, capable of simultaneously buffering predation,
3 thermal extremes, and invasive species pressure across a wide range of taxa and ecosystems
4 (Selwood & Zimmer, 2020). In Lake Rotoiti, catfish and thermal stratification — the two
5 primary drivers of kōura decline — cannot be managed at the lake scale: eradication is not
6 currently achievable, and targeted catfish removal in high density areas reduces local pressure
7 without eliminating the lake-wide population (Dedual, 2019). Emerging biotechnologies such
8 as gene drives and host-specific biological control agents may offer future options (Kopf et al.,
9 2019; Teem et al., 2020; Thresher et al., 2014), but these remain experimental, and thermal
10 stratification will continue to intensify under climate warming (Lee et al., 2025; Woolway et al.,
11 2021). Where the primary stressors cannot be removed, habitat addition becomes one of the
12 few available management tools, and refugia become correspondingly more valuable (Selwood
13 & Zimmer, 2020).

14 The stone piles deployed here were isolated structures, separated from each other and from
15 deeper water by soft sandy substrate. Isolated habitat patches support fewer species and are
16 more vulnerable to local extinction than connected habitats (Hanski, 1998), and the Winter
17 2026 decline is consistent with this limitation: stone piles lacked the connectivity to deep refuge
18 that allowed reference sites to remain stable. Orienting future structures perpendicular to the
19 shoreline, extending through macrophyte beds and into deeper water, could create corridors
20 linking the littoral zone with deeper refugia, allowing kōura to move more freely between depth
21 ranges as conditions change, as they do at reference sites. This design has not been tested
22 but is a promising direction for future deployments, with the addition of potential benefit of
23 suppressing macrophyte growth along the corridor.

24 The dense algal fouling observed in Winter 2026 also raises a placement question. In sheltered
25 bays, stone piles are protected from wave damage, but algal growth and sediment accumulation
26 could progressively reduce crevice accessibility and refugia value; in more exposed locations,
27 natural scouring could limit fouling, but conditions may be too disturbed to provide effective
28 shelter (Parkyn & Collier, 2004). This placement trade-off warrants direct investigation in
29 future deployments, and standardised photography of consistent stone faces each monitoring
30 visit would allow fouling development to be tracked alongside changes in kōura use.

31 Finally, the five stone pile structures deployed here cannot be expected to drive population-
32 level recovery across a lake of this size on their own. The spatial and temporal scale of the
33 ecosystem responses is typically greater than the scale of the interventions used to produce
34 them (Minns et al., 1996), and scaling from individual stone pile deployment to habitat addition

1 at population-relevant extents is a central challenge in restoration ecology ([Ngwenya et al.,](#)
2 [2025](#)).

3 Beyond the ecological results, this study establishes a practical foundation for scaling habitat
4 enhancement in these lakes: the approvals, consents, and construction methodology needed
5 to deploy stone piles have now been tested, with evidence that doing so is unlikely to have a
6 counterproductive result by benefiting catfish. These outcomes give lake managers a practical,
7 tested option for future large-scale deployment; continued monitoring for at least three years is
8 recommended to determine whether the Winter 2026 decline was a one-off event and whether
9 kōura abundance at stone pile sites continues to build in subsequent spring and summer
10 seasons.

11 **Conclusion**

12 Stone pile deployment alone cannot reverse the trajectory of kōura decline in Lake Rotoiti,
13 but it directly addresses the current bottleneck: the shortage of structural refuge in a lake
14 where soft sediment dominates the shorelines and where the two primary stressors, catfish
15 predation and thermal habitat compression, cannot be controlled at the lake scale. Both
16 stressors are expected to intensify under continued climate warming ([Lee et al., 2025](#); [Woolway](#)
17 [et al., 2021](#)), narrowing the band of habitat available to kōura and increasing, rather than
18 diminishing, the value of structural refugia over time. Establishing refugia before populations
19 decline further may be more effective than waiting until source populations fall below critical
20 densities for recolonisation ([Kusabs et al., 2026](#); [Morelli et al., 2016](#)). The consistent positive
21 directional trend in occupancy, the rapid and clearly seasonal use of the structures, and the
22 developing biological community on the stone pile surfaces together suggest the foundations for
23 a population-level response by kōura are being laid, even though catfish showed no comparable
24 benefit from the structures. Whether this translates into a detectable population-level effect
25 will depend on continued long-term monitoring — the logical next phase of this restoration
26 programme.

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13 **Authors' contributions**

14 Study conception and design were undertaken by Olivier V. Raven, Ian A. K. Kusabs, Robin
15 Holmes, Francis J. Burdon and Deniz Özkundakci. Fieldwork was executed by Olivier V. Raven.
16 Data analysis was performed by Olivier V. Raven. Ian A. K. Kusabs obtained council and iwi
17 consent and provided local ecological knowledge. Manuscript drafting was led by Olivier V.
18 Raven. All authors contributed to manuscript revision, approved the results, and the final
19 manuscript.

20 **Use of Artificial Intelligence**

21 Use of Artificial Intelligence: Claude (Anthropic Sonnet 5.0) was used to assist with manuscript
22 editing and language refinement. All content was critically reviewed, and the authors take full
23 responsibility for the final manuscript.

24 **Declarations**

25 The corresponding author has declared that none of the authors has any competing interests.

1 Data availability statement

2 The code and derived data supporting the findings of this study are openly available on GitHub
3 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
4 is hosted at https://olivierraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
5 archived on Zenodo (<https://doi.org/...>). Peer review documents, including the cover letter
6 and reviewer responses, are available in the manuscript repository in the interest of open and
7 transparent science.

8 Supplementary information

9 *Supplementary figures are provided in the appendices.*

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